

2025 Calculus Midterm

L.A. and Calculus

① a) $\lim_{x \rightarrow 0} \frac{1}{x^3}$. Notice that $\lim_{x \rightarrow 0^-} \frac{1}{x^3} = -\infty$ and $\lim_{x \rightarrow 0^+} \frac{1}{x^3} = +\infty$. Since $\lim_{x \rightarrow 0^-} \frac{1}{x^3} \neq \lim_{x \rightarrow 0^+} \frac{1}{x^3}$, the limit doesn't exist.

b) $\lim_{x \rightarrow \infty} \frac{5x^3 + x - 10}{2x^2 + 3x + 4} = \lim_{x \rightarrow \infty} \frac{5x + 1/x - 10/x^2}{2 + 3/x + 4/x^2} = \infty$.

② Let $f(x) = \frac{e^x - 1 - x}{x^2}$. Find $\lim_{x \rightarrow 0} f(x)$.

Close to 0 $e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots$, so $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots - 1 - x}{x^2} =$
 $= \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots}{x^2} = \lim_{x \rightarrow 0} \frac{1}{2} + \frac{1}{3!}x + \dots = \frac{1}{2}$.

③ Show that there exists a solution to $\cos x = x$ in $[0, 1]$.

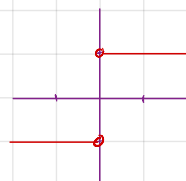
Solutions to $\cos x = x$ are the same as to $\cos x - x = 0$. The function $f(x) = \cos x - x$ is continuous, and $f(0) = \cos 0 - 0 = 1 > 0$ and $f(1) = \cos 1 - 1$. Since $\cos x \in [0, 1] \forall x \in \mathbb{R}$, $\cos 1 < 1$, thus, $f(1) = \cos 1 - 1 < 0$. By Bolzano's Theorem there is a zero of f in $[0, 1]$ and thus a solution to $\cos x = x$ in $[0, 1]$.

④ Let $f(x) = |x|$

a) Why f' doesn't exist at $x = 0$? Because the $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$ doesn't exist.

b) Find f' elsewhere and draw it.

Since $f(x) = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases} \rightarrow f'(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x > 0 \end{cases}$.



c) Find f'' elsewhere and draw it.

$f''(x) = 0, \forall x \in (-\infty, 0) \cup (0, \infty)$.
(or $\mathbb{R} \setminus \{0\}$)



⑤ $f(x) = \begin{cases} \frac{x^2-1}{x-1} & x \neq 1 \\ 2 & x = 1 \end{cases}$. Is f continuous?

For f to be continuous, the limit at 1 must coincide with the image. So $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{(x-1)} = \lim_{x \rightarrow 1} (x+1) = 2 = f(1)$. So yes, f is continuous.

⑥ $g(x, y, z) = x^3 + yz^2$, $x(t) = t^2$, $y(t) = e^t$, $z(t) = \sin t$. Compute $\frac{d}{dt} g(x(t), y(t), z(t))$.

Partial derivatives: $g_x = 3x^2$, $g_y = z^2$, $g_z = 2yz$.

Derivatives: $\frac{d}{dt} x(t) = 2t$, $\frac{d}{dt} y(t) = e^t$, $\frac{d}{dt} z(t) = \cos t$.

Apply chain rule: $\frac{d}{dt} g = 3x^2(2t) + z^2 \cdot e^t + 2yz \cdot \cos t$

Substitute: $\frac{d}{dt} g(t) = 3(t^2)^2 \cdot (2t) + \sin^2 t \cdot e^t + 2e^t \sin t \cos t = 6t^5 + e^t(\sin^2 t + 2\sin t \cos t)$